

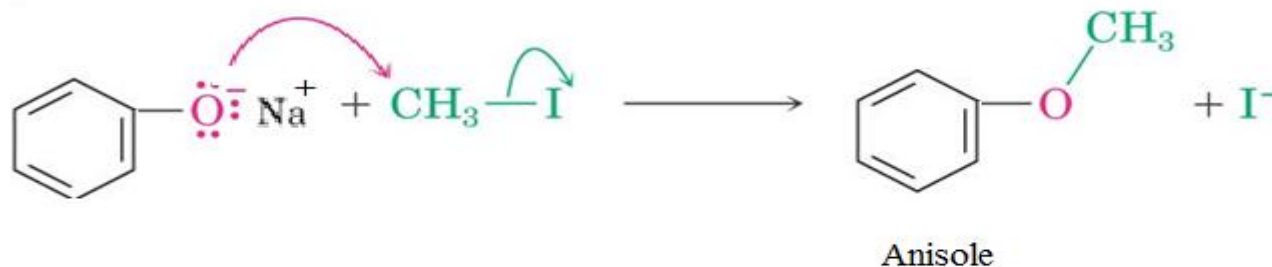
Aromatic Ethers

Aromatic ether can be divided into two groups

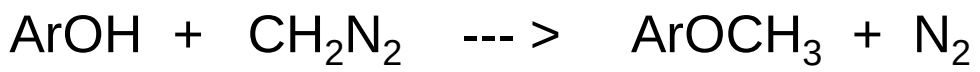
1. Phenolic ethers (alkaryl ethers, Ar –O-R)
2. Diaryl ethers (Ar -O- Ar)

Synthesis

1. Phenolic ethers (alkaryl ethers, Ar –O-R) by Williamson Ether Synthesis
These can be prepared by heating Sodium phenoxide with alkyl halide in ethanol solution or by treating an alkaline solution of a phenol with alkyl sulphate

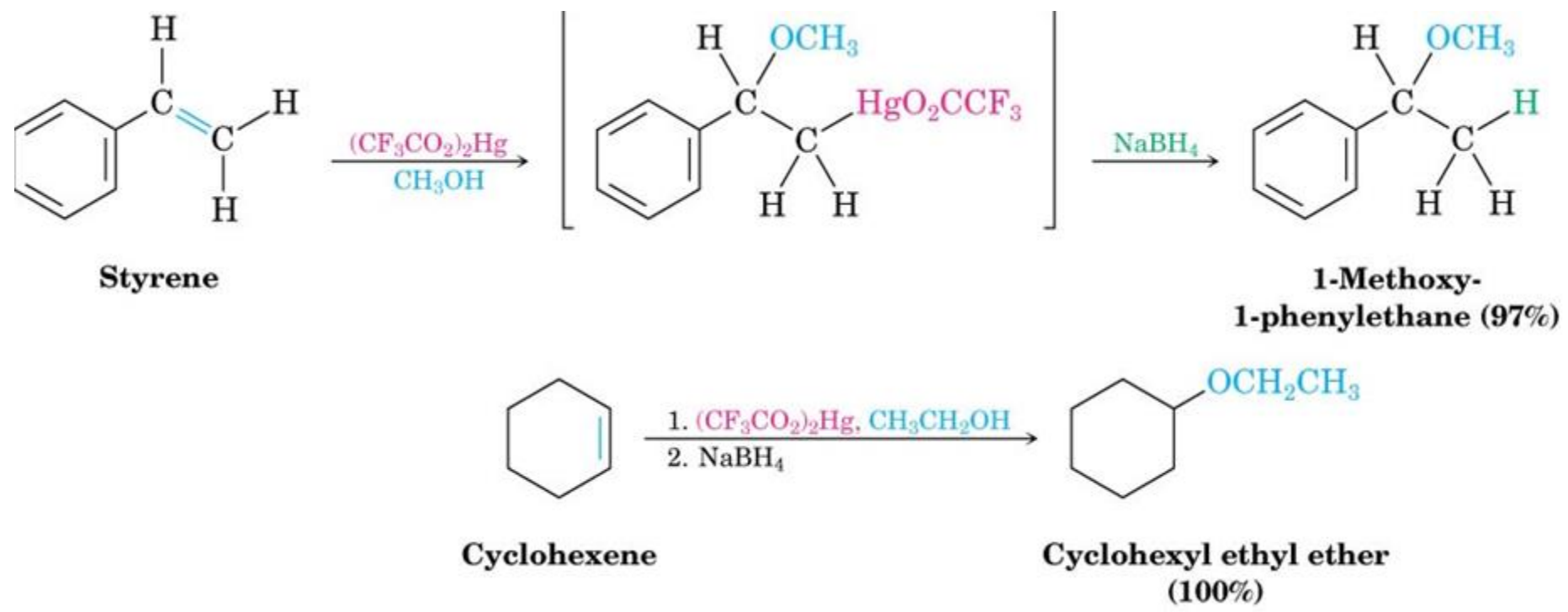


Phenolic methyl esters can be obtained by reacting diazomethane with phenol



Alkoxymercuration of Alkenes

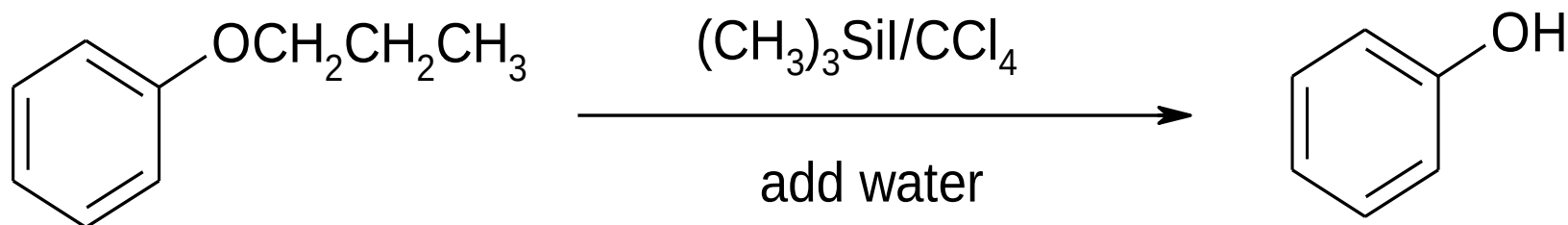
- Alkene + alcohol and mercuric trifluoroacetate
- Remove Hg with NaBH_4 yields an ether



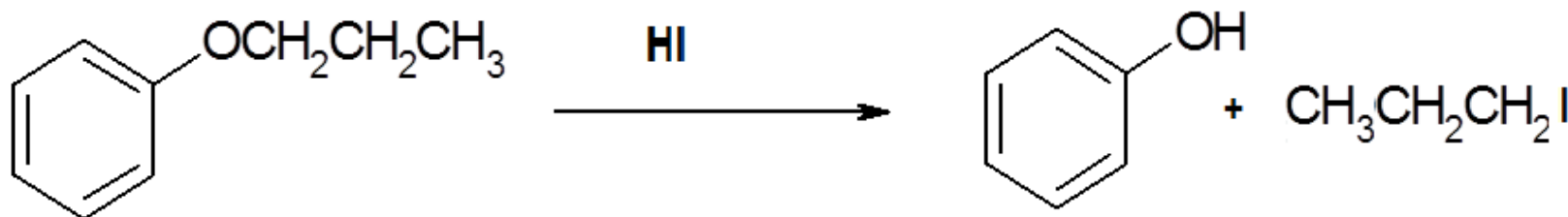
Reactions of Ethers

1. Cleavage with Trimethylsilyl Iodide

- More mild reagent developed



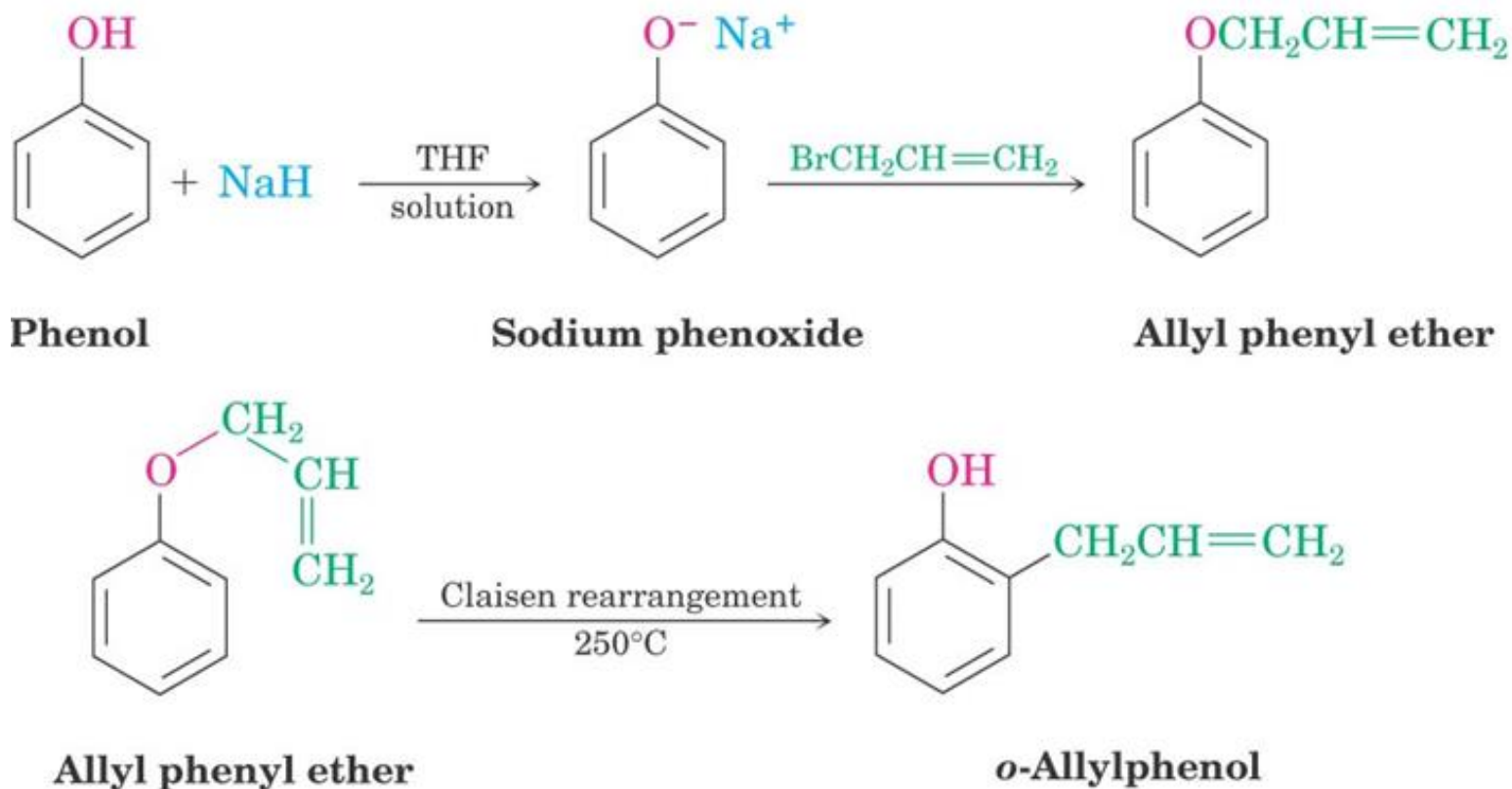
- Iodide reacts with Less hindered side



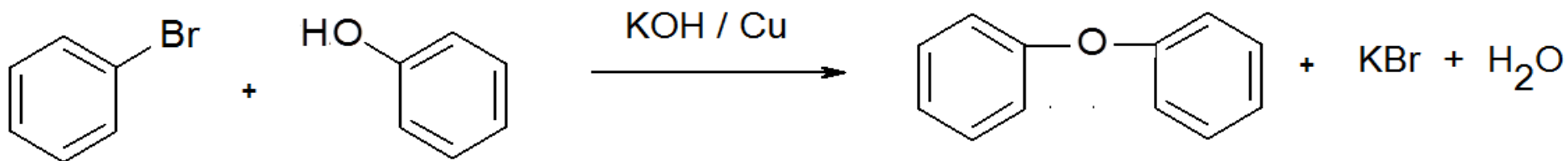
Allyl Aryl Ethers

The Claisen Rearrangement

- Rearrangement of allyl aryl ether ($\text{ArOCH}_2\text{CH}=\text{CH}_2$) into allylphenol
- Heating to 200–250°C leads to an *o*-allylphenol

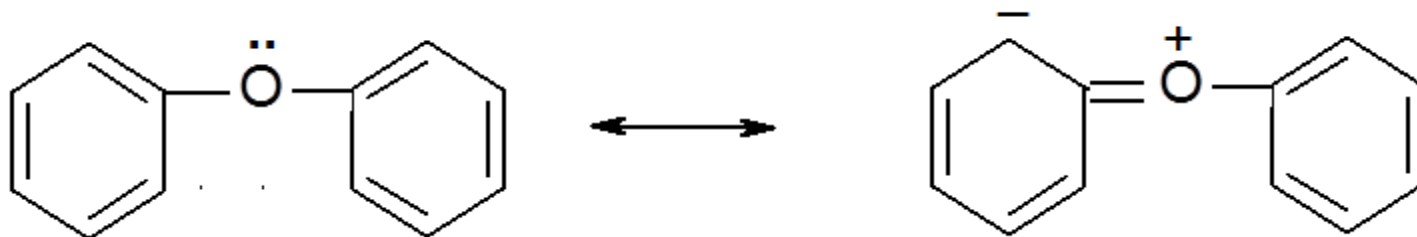


Diaryl ethers



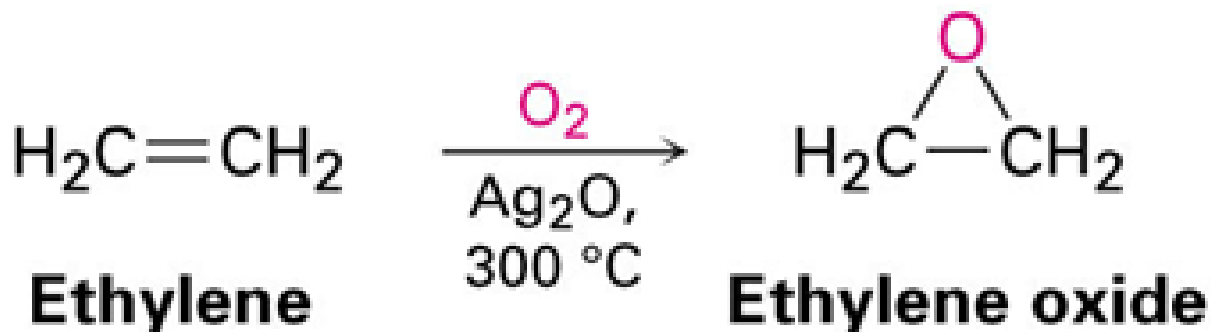
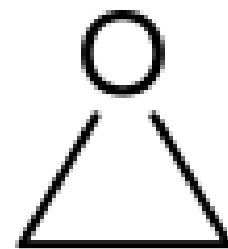
Diphenyl ether

Diaryl ethers are extremely weak bases, and this can be explained by resonance hybrids

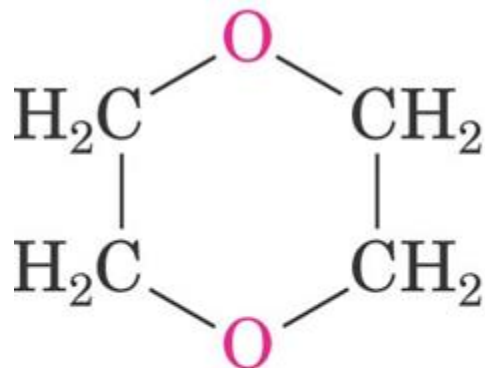


Cyclic Ethers: Epoxides

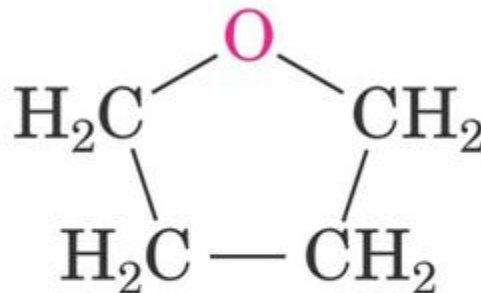
- Cyclic ethers behave like acyclic ethers,
- Three membered ring ether is called an oxirane Also called epoxides
- Ethylene oxide (oxirane; 1,2-epoxyethane) is industrially important as an intermediate
- Prepared by reaction of ethylene with oxygen at 300 °C and silver oxide catalyst



Dioxane and tetrahydrofuran are used as solvents



1,4-Dioxane



Tetrahydrofuran

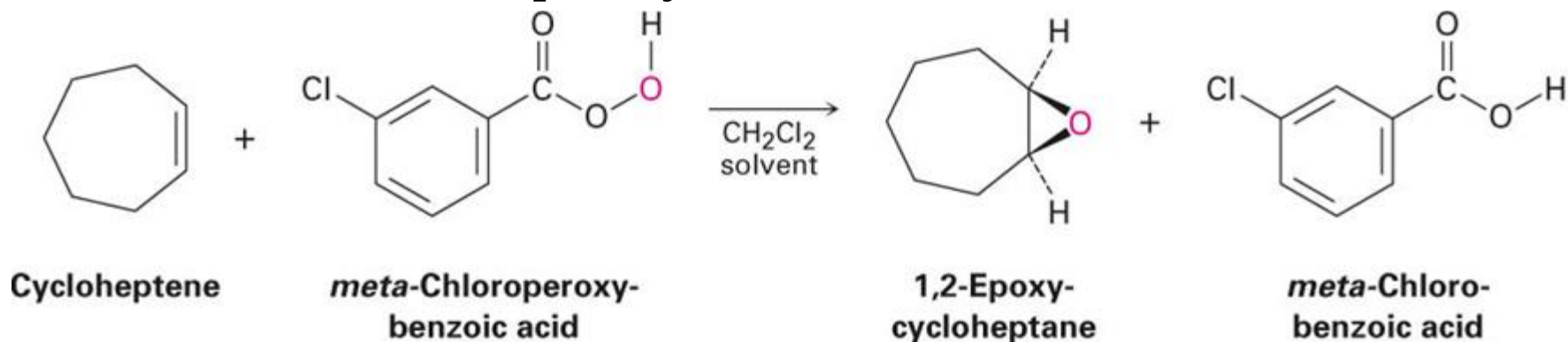
Synthesis

1. Epoxides from Halohydrins

- Addition of HO-X to an alkene gives a halohydrin
- Treatment of a halohydrin with base gives an epoxide



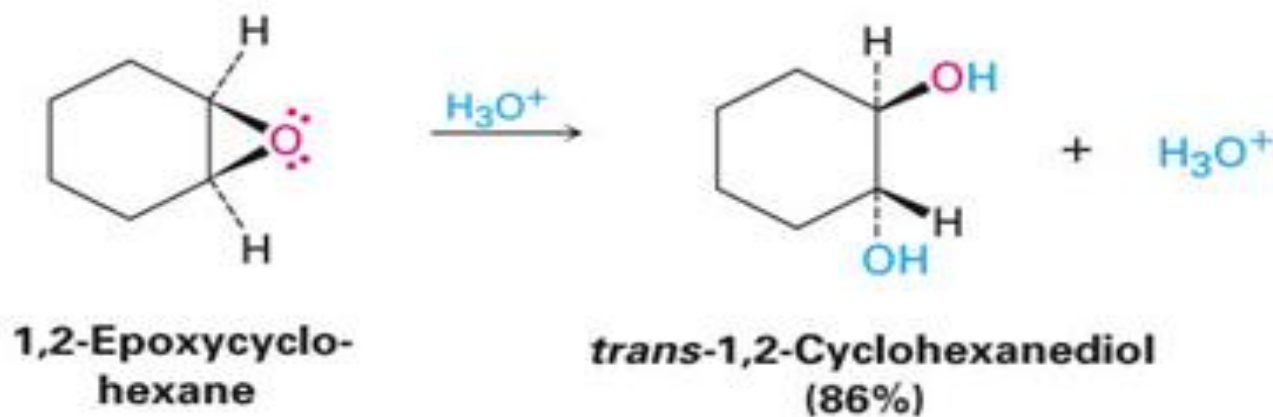
2. Treat an alkene with a peroxyacid



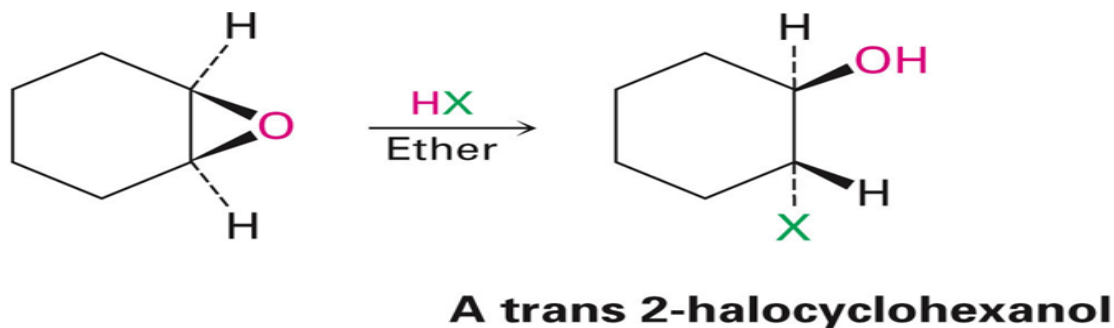
Reactions

Ring Opening

- Water adds to epoxides with dilute acid at room temperature
- Product is a 1,2-diol (on adjacent C's: *vicinal*)



Anhydrous HF, HBr, HCl, or HI combines with an epoxide gives trans product



where $\text{X} = \text{F, Br, Cl, or I}$